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Note

Total Synthesis of Isohericenone J via a Stille Coupling Reaction

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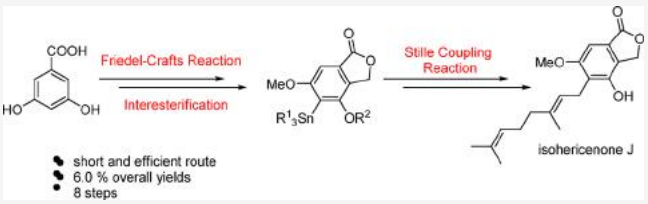
ABSTRACT: The first total synthesis of isohericenone J is reported. Key features of this synthetic strategy are a Friedel–Crafts reaction to construct the isobenzofuranone unit and a Pd-catalyzed Stille coupling reaction for the formation of the C5–C1′ bond, generating the natural product, as well as one of its isomers, in 6.0% overall yield in eight steps. This strategy provides a foundation for the synthesis of challenging isobenzofuranone and isoindolinone-type derivatives.



Isohericenone J, an isobenzofuranone derivative, was isolated from the mushroom *Hericium erinaceum*, and in 2015, it was determined by the Kim group to inhibit the growth of HL-60 and HEL-299 cells.[1] Previous investigations have shown that polysaccharides,[2] fatty acids,[3] diterpenoids,[4] alkaloids,[5] sterols,[6] and aromatic compounds such as isohericenone J,[1] hericenone A,[7,14] erinacerin B,[8] corallocins A,[9] and isohericerin[10] (Figure 1) can be isolated from lion's mane mushroom;

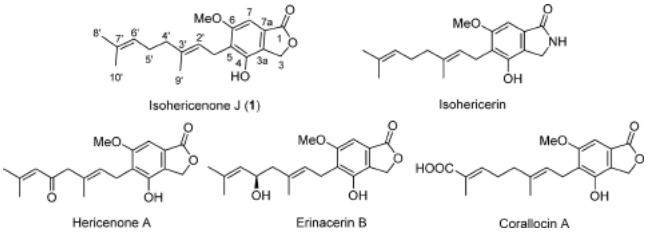
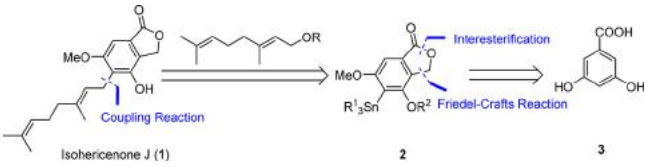


Figure 1. Structures of isohericenone J and related natural products.

this mushroom is consumed as a low-calorie food by humans and has been used in the treatment of gastritis in traditional Chinese medicine for more than 1000 years.[11] It is desirable to develop an efficient strategy to accomplish the total syntheses of all the isobenzofuranone-type natural products isolated from this mushroom. The efficient strategy will provide a foundation for the synthesis of challenging isoindolinone-type derivatives, such as chartarutines A–H and stachybotrysams A–D, prenylated isoindolinone derivatives, obtained from *Stachybotrys chartarum*.[12]

Herein, a short and efficient synthesis of isohericenone J and one of its isomers is disclosed. As shown in Figure 1, isohericenone J (**1**) features an isobenzofuranone skeleton linked to a geranyl fragment. Based on this structural analysis, we realized that achieving overall synthetic efficiency would depend on (a) the simple and efficient construction of the phthalide unit and (b) a highly selective coupling of the geraniol derivative with the lactone segment. The synthesis of an isobenzofuranone with substituents at the C7a- and C3a-positions was reported in 1992 by the Rao group,[13] who revised the structure reported for hericenone A. Many synthetic strategies to construct the phthalide intermediate have been developed.[14] Even so, we committed ourselves to complete the formation of these natural products possessing a geranyl side chain attached to a phthalide skeleton in a simple and practical way. Our retrosynthetic analysis of isohericenone J (**1**) is depicted in Scheme 1.

Scheme 1. Retrosynthetic Analysis of Isohericenone J (1)



Initially, we considered isohericenone J (**1**) could be obtained from protected geraniol and trimethyltin (**2**) with the lactone core under Stille coupling conditions. The phthalide fragment in **2** was envisaged to be obtained from a Friedel–Crafts reaction, followed by an interesterification via a benzyl alcohol at C3a with the appropriate methyl ester, which would be easily obtained from benzoic acid (**3**).

Based on the above analysis, our investigation started with 3,5-dihydroxybenzoic acid (**3**, Scheme 2). The esterification of **3** followed by iodination with N-iodosuccinimide (NIS) in MeOH at 0 °C generated iodide **5** in 86% yield over two steps;

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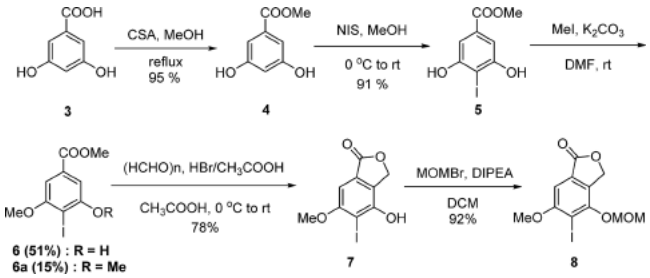
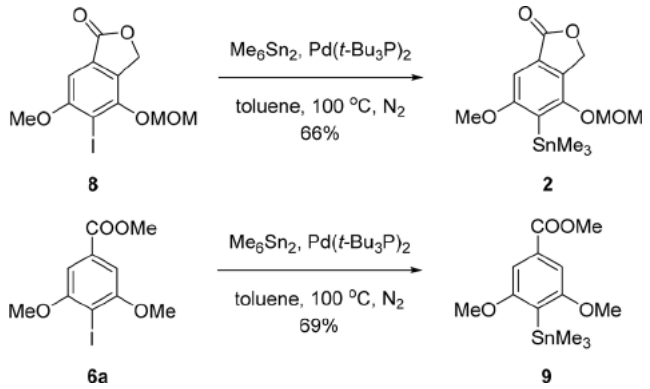
Scheme 2. Synthesis of Lactone 8



this compound is also commercially available. Treatment of diol **5** with iodomethane in the presence of K_2CO_3 in DMF as solvent provided monomethoxy-substituted product **6** and dimethoxy-substituted ester **6a** in yields of 51% and 15%, respectively. The desired lactone **7** was obtained in 78% yield via a Friedel−Crafts alkylation reaction and interesterification domino reaction sequence when using paraformaldehyde and HBr in acetic acid. Protecting the phenol was beneficial for the Stille coupling, so bromo(methoxy)methane was employed in CH_2Cl_2 in the presence of DIPEA to furnish protected intermediate **8** in 92% yield.

With lactone **8** in hand, the syntheses of two tin reagents, **2** and **9**, were addressed. The aryl iodides (**8** and **6a**) were converted to the corresponding aryl stannanes (**2** and **9**) by treatment with Me_6Sn_2 and $Pd(t\text{-}Bu_3P)_2$ in dry toluene at 100 °C under nitrogen in 66% and 69% yields, respectively (Scheme 3).

Scheme 3. Syntheses of Tin Reagents 2 and 9

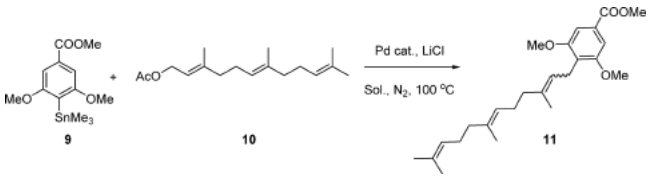


According to the coupling work of Hegedus,[15] the reaction conditions of the Stille coupling reaction need optimizing, and coupling reaction conditions using tin reagent **9** with acetylated (*E*,*E*)-farnesol (**10**) as a model substrate were evaluated.

Frequently used palladium catalysts such as $Pd(PPh_3)_4$ and $Pd(Ph_3P)_2Cl_2$ were tested first (Table 1, entries 1−5). However, only $Pd_2(dba)_3$ effectively catalyzed the reaction, and it gave coupling product **11** in 54% yield in the presence of LiCl with dry DMF as the solvent (entry 5). It has been shown that DMF as the organic solvent cannot be replaced by toluene (entry 6). In addition, the effect of the equivalents of the stannane was studied (entry 7). Compared to 1.0 equiv of coupling partner **9**, excess tin reagent, e.g., 2.0 equiv, improved the yield.

To demonstrate the feasibility of the coupling reaction, side chain **10** and lactone **2** were coupled (Table 2). Subjecting

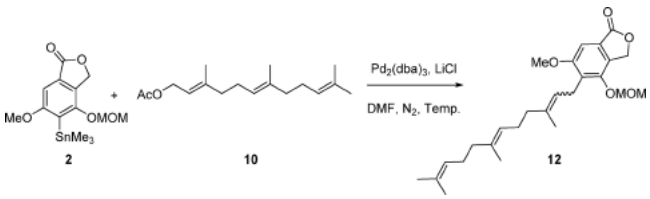
Table 1. Optimization of the Stille reaction conditions[a,b]



entry	equiv (**9**)	Pd (cat.)	solvent	yield (%)
1	1	$Pd(t\text{-}Bu_3P)_2$	DMF	0
2	1	$Pd(PPh_3)_4$	DMF	0
3	1	$Pd(OAc)_2$	DMF	0
4	1	$Pd(Ph_3P)_2Cl_2$	DMF	0
5[c]	1	$Pd_2(dba)_3$	DMF	54
6	1	$Pd_2(dba)_3$	toluene	0
7[d]	2	$Pd_2(dba)_3$	DMF	71

[a]Unless noted otherwise, all reactions were conducted using 0.1 mmol of **10**, 10% mmol of Pd cat., and 3.0 equiv of LiCl in 2.0 mL of solvent. [b]Isolated yield. [c]Formation of **11** [*E*/*Z* (1.2:1)]. [d]Formation of **11** [*E*/*Z* (1.2:1)].

Table 2. Optimization of the Stille Reaction Conditions[a,b]



entry	equiv (**2**)	temp (°C)	yield (%)	ratio (*E*/*Z*)
1	1	100	27	1.2:1
2	1.5	100	36	1.2:1
3	1.5	50	37	1.2:1
4	3	50	73	1.1:1
5	3	rt	0	

[a]Unless noted otherwise, all reactions were conducted using 0.1 mmol of **10**, 10% mmol of Pd cat., and 3.0 equiv of LiCl in 2.0 mL of solvent. [b]Isolated yield.

intermediate **2**, possessing the phthalide fragment, to these conditions gave the target product **12** in 27% yield, as shown in Table 2 entry 1. Inspired by the result in entry 7 in Table 1, increased loading of **2** was investigated (Table 2, entry 2) at 100 °C, and the yield of coupling product **12** was increased to 36%. Next, the temperature was reduced to 50 °C, and this did not affect the reaction efficiency. A yield of 37% was not satisfactory to complete the total syntheses of isohericenone J and other natural products. Therefore, 3 equiv of substrate **2** was employed at 50 °C, and this furnished **12** in a higher yield (73%). This Stille coupling of these two side chains (**13** and **15**) could not be achieved at room temperature (entry 5).

With the optimized coupling reaction conditions, the target natural product, isohericenone J, was synthesized in 29% yield over two steps via a Stille coupling reaction in the presence of LiCl and deprotection with D-camphorsulfonic acid (CSA),[14c] and an isomer was obtained in 27% yield; these products were isolated by HPLC. The key Stille coupling was also studied with (*Z*)-3,7-dimethylocta-2,6-dien-1-yl acetate (**15**) under the same reaction conditions, and as expected, both *E*- and *Z*-products were observed, in an *E*/*Z* ratio of 2:3 (Scheme 4).

To confirm the structure of the isomer, the NOE spectra of the isolated products were acquired along with ^1H and ^{13}C

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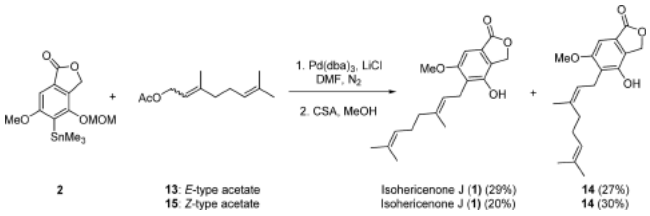
Scheme 4. Total Syntheses of Isohericeone J (1) and an Isomer (14)



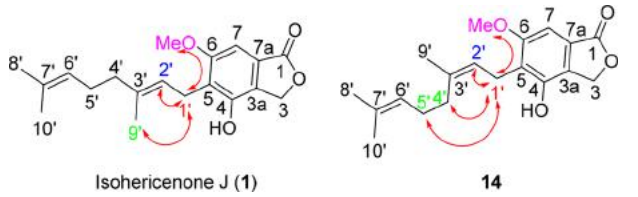


Figure 2. Key NOE correlations of isohericenone J and an isomer.

NMR spectra. The NOE correlations of the H of the methoxy group with H−C1′, of H−C1′ with H−C2′, and especially of H−C1′with H−C9′ assigned the *E* configuration of isohericenone J. At the same time, there were no significant differences in the ^{1}H and ^{13}C NMR chemical shifts between the final synthetic product and isohericenone J isolated by Kim and co-workers. However, the correlations of H−C1′ with H−C4′ and of H−C1′ with H−C5′ allow the configuration of the isomer to be determined.

In summary, a Friedel−Crafts alkylation reaction and a coupling reaction were the key steps in achieving total syntheses of isohericenone J and its isomer. If commercially available intermediate **5** was used as the starting material, the yield of the natural product was increased to 7.0% overall in six steps. This total synthetic synthesis strategy can be used as a guide to access other natural products with similar skeletons.[16,17]

■ EXPERIMENTAL SECTION

General Experimental Procedures. Unless otherwise mentioned, all reactions were performed in flame-dried glassware under air. Solvents were distilled prior to use. Reagents were used as purchased from J&K, Beijing Ouhe, Aldrich, Adamas-beta, Alfa Aesar, or TCI unless otherwise noted. Chromatographic separations were performed using Kangbino 48−75 Å SiO_2. ^{1}H and ^{13}C NMR spectra were recorded on Agilent Pro Pulse spectrometers using $CDCl_3$ with TMS or residual solvent as standard unless otherwise noted. Melting points were determined using a Laboratory Device WRS-3 and are uncorrected/calibrated. TLC analysis was performed using Kangbino glass-backed plates (60 Å, 250 μm) and visualized using UV and phosphomolybdic acid stains. Low-resolution mass spectra were obtained using an Agilent 1100 series LS/MSD. High-resolution mass spectra for new compounds were recorded on a Thermo Fisher Exactive APCI-orbitrap spectrometer or an Agilent 1290 Infinity II-6530B. All spectral data obtained for new compounds are reported here.

Synthesis of Methyl 3,5-Dihydroxybenzoate, 4. To a solution of 3,5-dihydroxybenzoic acid **3** (1.54 g, 10.00 mmol) in MeOH (20 mL) was added CSA (232 mg, 1.00 mmol) at room temperature. The solution was refluxed until all starting material was consumed (TLC monitoring). Saturated $NaHCO_3$ solution was added. The aqueous layer was extracted with CH_2Cl_2; then the solvent was removed in vacuo, and the residue was purified by silica gel column chromatography eluting with 40% ethyl acetate in petroleum ether to afford methyl 3,5-dihydroxybenzoate (**4**, 1.60 g).

4: 95%, white solid, mp: 170−172 °C; R_f = 0.2 (50% EtOAc/petroleum ether); ^{1}H NMR (400 MHz, methanol-d_4) δ 6.92 (d, J = 2.3 Hz, 2H), 6.47 (td, J = 2.3, 0.6 Hz, 1H), 3.85 (s, 3H); ^{13}C NMR (100 MHz, methanol-d_4) δ 168.7, 159.8, 133.0, 108.8, 108.2, 52.5; HRMS-ESI (m/z) calcd for $C_8H_7O_4$ (M − H)$^-$ 167.0350, found 167.0354.

Synthesis of Methyl 3,5-Dihydroxy-4-iodobenzoate, 5. To a solution of **4** (1.50 g, 8.92 mmol) in MeOH (10 mL) was added slowly a solution of NIS (2.11g, 9.37 mmol) in MeOH (10 mL) at 0 °C. The reaction mixture was slowly warmed to room temperature while mixing. After 16 h the reaction mixture was diluted with ice water and quenched by saturated Na_2SO_3, then extracted with EtOAc. The combined organic layers were washed with brine, dried over Na_2SO_4, and concentrated in vacuo. After purification by flash column chromatography (petroleum ether/EtOAc, 60:40), the title compound **5** was obtained in a yield of 2.39 g.

5: 91%, white solid, mp 200−202 °C; R_f = 0.4 (50% EtOAc/petroleum ether); ^{1}H NMR (400 MHz, methanol-d_4) δ 6.96 (s, 2H), 3.85 (s, 3H); ^{13}C NMR (100 MHz, methanol-d_4) δ 168.3, 159.6, 132.5, 107.2, 82.0, 52.6; HRMS-ESI (m/z) calcd for $C_8H_6IO_4$ (M − H)$^-$ 292.9316, found 292.9315.

Synthesis of Methyl 3-Hydroxy-4-iodo-5-methoxybenzoate, 6, and Methyl 4-Iodo-3,5-dimethoxybenzoate, 6a. Methyl 3,5-dihydroxy-4-iodobenzoate (2.00 g, 6.80 mmol) and K_2CO_3 (1.88 g, 13.60 mmol) were dissolved in DMF (15 mL). Then iodomethane (0.43 mL, 6.80 mmol) in DMF (15 mL) was added dropwise and stirred at room temperature for 40 h. NH_4Cl (saturated, aqueous) was added, and the mixture was extracted with EtOAc. The organic layers were washed with brine, dried over Na_2SO_4, filtered, and concentrated. Purification was achieved using column chromatography on silica gel (petroleum ether/EtOAc, 90:10−60:40) to give methyl 3-hydroxy-4-iodo-5-methoxybenzoate (**6**, 1.07 g) and byproduct **6a** (329 mg).

6: 51%, white solid; mp 128−130 °C; R_f = 0.7 (50% EtOAc/petroleum ether); ^{1}H NMR (400 MHz, methanol-d_4) δ 7.13−7.11 (m, 1H), 7.03−6.99 (m, 1H), 3.88 (s, 6H); ^{13}C NMR (100 MHz, methanol-d_4) δ 168.1, 161.2, 159.6, 132.8, 109.4, 103.4, 83.3, 57.0, 52.8; HRMS-ESI (m/z) calcd for $C_9H_8IO_4$ (M − H)$^-$ 306.9473, found 306.9476.

6a: 15%, white solid; mp 114−115 °C; R_f = 0.8 (50% EtOAc/petroleum ether); ^{1}H NMR (400 MHz, chloroform-d) δ 7.14 (s, 2H), 3.93 (s, 6H), 3.91 (s, 3H); ^{13}C NMR (100 MHz, chloroform-d) δ 166.7, 159.5, 131.9, 104.8, 84.3, 56.9, 52.6; HRMS-ESI (m/z) calcd for $C_{10}H_{10}IO_4$ (M − H)$^-$ 320.9627, found 320.9623.

Synthesis of 4-Hydroxy-5-iodo-6-methoxyisobenzofuran-1(3*H*)-one, 7. To a stirred mixture of 1.00 g (3.25 mmol) of methyl 3-hydroxy-4-iodo-5-methoxybenzoate (**6**) and 586 mg (6.50 mmol) of paraformaldehyde in glacial acetic acid (5 mL) cooled to 0 °C was added slowly 0.81 mL (3.58 mmol) of HBr/glacial acetic acid (33%). Then the cooling bath was removed and the mixture was stirred at room temperature for 16 h. $NaHCO_3$ (saturated, aqueous) was added, and the mixture was extracted with EtOAc. The organic layers were dried over Na_2SO_4 and concentrated. Purification was achieved using column chromatography on silica gel (petroleum ether/EtOAc, 90:10−50:50) to give 4-hydroxy-5-iodo-6-methoxyisobenzofuran-1(3*H*)-one (**7**, 776 mg).

7: 78%, white solid; mp 180−182 °C; 78%; R_f = 0.3 (50% EtOAc/petroleum ether); ^{1}H NMR (400 MHz, $CDCl_3$) δ 6.89 (s, 1H), 5.89 (s, 1H), 5.27 (s, 2H), 3.96 (s, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 170.8, 160.1, 150.8, 128.6, 125.3, 98.4, 85.9, 68.0, 57.3; HRMS-ESI (m/z) calcd for $C_9H_6IO_4$ (M − H)$^-$ 304.9316, found 304.9322.

Synthesis of 5-Iodo-6-methoxy-4-(methoxymethoxy)-isobenzofuran-1(3*H*)-one, 8. 4-Hydroxy-5-iodo-6-methoxyisobenzofuran-1(3*H*)-one (**7**; 612 mg, 2.00 mmol) and DIPEA (0.66 mL, 4.00 mmol) were dissolved in CH_2Cl_2 (15 mL) at 0 °C. Then bromomethyl methyl ether (0.25 mL, 3.00 mmol) in CH_2Cl_2 (5 mL) was added dropwise and stirred at room temperature overnight. H_2O was added, and the mixture was extracted with CH_2Cl_2. The organic layers were dried over Na_2SO_4 and concentrated. Purification was achieved using column chromatography on silica gel (petroleum

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ether/EtOAc, 70:30) to give 5-iodo-6-methoxy-4-(methoxymethoxy)-isobenzofuran-1(3*H*)-one (**8**, 644 mg).

8: 92%, white solid; mp 163−165 °C; R_f = 0.6 (40% EtOAc/petroleum ether); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.07 (s, 1H), 5.40 (s, 2H), 5.17 (s, 2H), 3.97 (s, 3H), 3.58 (s, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 170.6, 160.8, 152.7, 128.7, 101.4, 97.6, 91.7, 77.4, 68.6, 57.4, 57.3; HRMS-ESI (*m*/*z*) calcd for $C_{11}H_{12}IO_5$ (M + H)$^+$ 350.9724, found 350.9722.

Synthesis of 6-Methoxy-4-(methoxymethoxy)-5-(trimethylstannyl)isobenzofuran-1(3*H*)-one, 2, and Methyl 3,5-Dimethoxy-4-(trimethylstannyl)benzoate, 9. 5-Iodo-6-methoxy-4-(methoxymethoxy)isobenzofuran-1(3*H*)-one (**8**; 640 mg, 1.83 mmol), bis(tri-*tert*-butylphosphine)palladium(0) (46.8 mg, 0.09 mmol), and hexamethyldistannane (1.56 mL, 2.75 mmol) were dissolved in toluene (20 mL) at room temperature under nitrogen. The mixture was stirred at 100 °C for 24 h. H_2O was added, and the mixture was extracted with CH_2Cl_2. The organic layers were washed with a saturated potassium fluoride solution in 10% NH_4OH prior to being dried to remove tin-containing residuals. The organic layers were dried over Na_2SO_4 and concentrated. Purification was achieved using column chromatography on silica gel (petroleum ether/EtOAc, 95:5) to give 6-methoxy-4-(methoxymethoxy)-5-(trimethylstannyl)-isobenzofuran-1(3*H*)-one (**2**, 469 mg).

2: 66%, white solid; mp 93−94 °C; R_f = 0.5 (20% EtOAc/petroleum ether); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.05 (s, 1H), 5.38 (s, 2H), 5.04 (s, 2H), 3.82 (s, 3H), 3.50 (s, 3H), 0.33 (s, 9H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 171.3, 166.1, 156.9, 130.3, 129.4, 127.0, 100.5, 96.2, 69.6, 56.8, 55.9, −6.8; HRMS-ESI (*m*/*z*) calcd for $C_{14}H_{21}O_5Sn$ (M + H)$^+$ 389.0406, found 389.0402.

9: 69%, white solid; mp 138−139 °C; R_f = 0.5 (40% EtOAc/petroleum ether); ^{1}H NMR (400 MHz, chloroform-*d*) δ 7.16 (s, 2H), 3.92 (s, 3H), 3.80 (s, 6H), 0.44−0.16 (m, 10H); ^{13}C NMR (100 MHz, chloroform-*d*) δ 167.4, 165.1, 133.0, 123.7, 104.2, 55.7, 52.3, −7.1.

Synthesis of (*E*)-3,7-Dimethylocta-2,6-dien-1-yl acetate, 13, (*Z*)-3,7-Dimethylocta-2,6-dien-1-yl Acetate, 15, and (2*E*,6*E*)-3,7,11-Trimethyldodeca-2,6,10-trien-1-yl Acetate, 10. (*E*)-3,7-Dimethylocta-2,6-dien-1-ol (1.00 g, 6.50 mmol), DMAP (15.8 mg, 0.13 mmol), and TEA (1.08 mL, 7.80 mmol) were added, and the resulting solution was cooled to 0 °C. Acetyl chloride was added dropwise, and the reaction stirred for 3 h. The reaction mixture was acidified to pH 2−3 with HCl (1 N). The organic layers were dried over Na_2SO_4 and concentrated. Purification was achieved using column chromatography on silica gel (petroleum ether/EtOAc, 97:3) to give (*E*)-3,7-dimethylocta-2,6-dien-1-yl acetate (**13**, 1.19 g).

13: 94%, colorless oil; R_f = 0.7 (10% EtOAc/petroleum ether); ^{1}H NMR (400 MHz, chloroform-*d*) δ 5.32 (t, *J* = 7.1 Hz, 1H), 5.06 (t, *J* = 7.1 Hz, 1H), 4.56 (d, *J* = 7.1 Hz, 2H), 2.12−2.05 (m, 2H), 2.05−1.98 (m, 2H), 2.03 (s, 3H), 1.68 (s, 3H), 1.66 (s, 3H), 1.58 (s, 3H); ^{13}C NMR (100 MHz, chloroform-*d*) δ 171.2, 142.4, 131.9, 123.8, 118.3, 61.5, 39.6, 26.3, 25.8, 21.1, 17.8, 16.5; HRMS-ESI (*m*/*z*) calcd for $C_{12}H_{20}O_2Na$ (M + Na)$^+$ 219.1356, found 219.1349.

15: 95%, colorless oil; R_f = 0.7 (10% EtOAc/petroleum ether); ^{1}H NMR (400 MHz, chloroform-*d*) δ 5.36 (t, *J* = 7.1 Hz, 1H), 5.13−5.05 (m, 1H), 4.56 (d, *J* = 7.1 Hz, 2H), 2.15−2.07 (m, 4H), 2.05 (s, 3H), 1.77 (s, 3H), 1.68 (s, 3H), 1.60 (s, 3H); ^{13}C NMR (100 MHz, chloroform-*d*) δ 171.3, 142.8, 132.3, 123.7, 119.2, 61.2, 32.3, 26.7, 25.8, 23.7, 21.2, 17.8.

10: 93%,colorless oil; R_f = 0.7 (10% EtOAc/petroleum ether); ^{1}H NMR (400 MHz, DMSO-d_6) δ 5.31−5.25 (m, 1H), 5.13−5.01 (m, 2H), 4.51 (d, *J* = 7.1 Hz, 2H), 2.11−1.99 (m, 5H), 1.99−1.96 (m, 3H), 1.96−1.89 (m, 2H), 1.66 (s, 3H), 1.63 (s, 3H), 1.56 (s, 6H); ^{13}C NMR (100 MHz, DMSO-d_6) δ 170.2, 141.2, 134.7, 130.6, 124.1, 123.5, 118.6, 60.5, 39.2, 38.9, 26.2, 25.6, 25.5, 20.7, 17.5, 16.1, 15.8; HRMS-ESI (*m*/*z*) calcd for $C_{17}H_{29}O_2$ (M + H)$^+$ 265.2162, found 265.2157.

Synthesis of Isohericenone J and Its Isomer and Compounds 11 and 12. To a mixture of 6-methoxy-4-(methoxymethoxy)-5-(trimethylstannyl)isobenzofuran-1(3*H*)-one **2** (116 mg, 0.3 mmol), (*E*)-3,7-dimethylocta-2,6-dien-1-yl acetate (**13**; 19.6 mg, 0.1 mmol), and lithium chloride (12.7 mg, 0.3 mmol) in DMF (2 mL) was added Pd(dba)$_3$ (3 mg, 0.005 mmol, 5 mol %) under nitrogen. The mixture was stirred at 50 °C for 24 h and then partitioned between water and CH_2Cl_2. H_2O was added, and the mixture was extracted with CH_2Cl_2. The organic layers were washed with brine and a saturated potassium fluoride solution in 10% NH_4OH prior to being dried to remove tin-containing residues. The organic layers were dried over Na_2SO_4 and concentrated. Purification was achieved using column chromatography on silica gel (petroleum ether/EtOAc, 95:5) to give the coupling product (24 mg).

To a solution of coupling product (24 mg, 0.06 mol) in MeOH (5 mL) was added CSA (33 mg, 0.14 mmol). The mixture was stirred at room temperature for 70 h and quenched by the addition of TEA. The mixture was extracted with CH_2Cl_2. The organic layers were dried over Na_2SO_4 and concentrated. Purification was achieved using column chromatography on silica gel (petroleum ether/EtOAc, 85:15) to give a mixture of isohericenone J and its isomer (17 mg). The mixture was further purified by HPLC using MeOH/H_2O (70:30, 3.0 mL/min) to give isohericenone J (8.1 mg, t_R = 31.0 min) and its isomer (7.6 mg, t_R = 32.0 min).

Isohericenone J: 29%, white solid; mp 95−97 °C; R_f = 0.4 (40% EtOAc/petroleum ether); ^{1}H NMR (400 MHz, chloroform-*d*) δ 6.97 (s, 1H), 6.04 (s, 1H), 5.25 (t, *J* = 7.1 Hz, 1H), 5.21 (s, 2H), 5.03−4.99(m, 1H), 3.87 (s, 3H), 3.48 (d, *J* = 7.1 Hz, 1H), 2.17−2.07 (m, 4H), 1.82 (s, 3H), 1.67 (s, 3H), 1.60 (s, 3H); ^{13}C NMR (100 MHz, chloroform-*d*) δ 171.8, 159.2, 150.6, 141.1, 132.7, 127.4, 125.3, 123.6, 121.1, 120.4, 98.6, 68.1, 56.4, 39.8, 26.3, 25.8, 23.1, 17.9, 16.4; HRMS-ESI (*m*/*z*) calcd for $C_{19}H_{23}O_4$ (M − H)$^-$ 315.1602, found 315.1602.

Isomer 14: 27%, white solid; mp 56−57 °C; R_f = 0.4 (40% EtOAc/petroleum ether); ^{1}H NMR (500 MHz, chloroform-*d*) δ 6.97 (s, 1H), 5.95 (s, 1H), 5.24 (t, *J* = 6.7 Hz, 1H), 5.21 (s, 2H), 5.14 (t, *J* = 6.7 Hz, 1H), 3.87 (s, 3H), 3.51 (d, *J* = 6.7 Hz, 2H), 2.30−2.23 (m, 2H), 2.21−2.14 (m, 2H), 1.79 (s, 3H), 1.71 (s, 3H), 1.64 (s, 3H); ^{13}C NMR (125 MHz, chloroform-*d*) δ 171.8, 159.2, 150.4, 141.1, 133.1, 130.2, 127.2, 125.3, 123.5, 121.1, 98.6, 68.1, 56.4, 32.2, 26.3, 25.9, 23.6, 22.8, 17.9; HRMS-ESI (*m*/*z*) calcd for $C_{19}H_{23}O_4$ (M − H)$^-$ 315.1602, found 315.1603.

11 (mixtures): colorless oil; R_f = 0.3 (20% EtOAc/petroleum ether); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.23 (s, 2H), 5.24−5.03 (m, 3H), 3.91 (s, 3H), 3.86 (s, 6H), 3.43−3.35 (m, 2H), 2.29−1.87 (m, 8H), 1.79−1.53 (m, 12H).

12 (mixtures): colorless oil; R_f = 0.2 (10% EtOAc/petroleum ether); ^{1}H NMR (400 MHz, chloroform-*d*) δ 7.13 (s, 1H), 5.38−5.36 (m, 2H), 5.20−5.09 (m, 2H), 5.08−5.04 (m, 3H), 3.89−3.87 (m, 3H), 3.54−3.52 (m, 3H), 3.48−3.41 (m, 2H), 2.16−1.87 (m, 8H), 1.70−1.55 (m, 12H).

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jnatprod.9b01251.

^{1}H, ^{13}C, and NOE NMR spectra of all compounds (PDF)

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Notes

The authors declare no competing financial interest.

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